JOMHACK
EMPOWERING INNOVATION

```
> bootcamp.learn()
```

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JomCode Coding Bootcamp

CHAPTER 04

Comparisons, Boolean logic and conditions

`learn / type / run / explain`

Plain meaning and technical explanation

PLAIN MEANING

A program chooses what to do when a condition is true or false.

TECHNICAL EXPLANATION

Comparisons produce bool values. if/elif/else selects a branch. and, or and not combine conditions. Indentation defines blocks.

REAL-WORLD USE

Reject blank titles and allow authors to edit their own posts.

QUICK TIP

Use == to compare values. A single = assigns a value.

INSTALL / RUN

No packages. Windows: py workbook.py. macOS: python3 workbook.py.

Worked example 1: Title length rules

workbook.py

```
01    title = "  Hello  ".strip()
02    valid = 1 <= len(title) <= 160
03    if valid:
04        print("Ready")
05    else:
06        print("Check title")
```

PRO TIP Clean before measuring.

Run and explain example 1

EXPECTED RESULT

Ready

WHY IT BEHAVES THIS WAY

The cleaned title has five characters, so the chained comparison is true.

Worked example 2: Choose one status

workbook.py

```
views = 25
if views == 0:
    label = "New"
elif views < 100:
    label = "Growing"
else:
    label = "Popular"
print(label)
```

PRO TIP Put specific cases before broader cases.

Run and explain example 2

EXPECTED RESULT

Growing

WHY IT BEHAVES THIS WAY

Only the first matching branch runs.

Worked example 3: Ownership decision

workbook.py

```
signed_in = True
user_id = "user-a"
author_id = "user-b"
can_edit = signed_in and user_id == author_id
print(can_edit)
print(not signed_in)
```

PRO TIP A hidden edit button is only a UI rule; the server must also check ownership.

Run and explain example 3

EXPECTED RESULT

False
False

WHY IT BEHAVES THIS WAY

Both parts of an and expression must be true. The author IDs differ.

Worked example 4: Missing and empty values

workbook.py

```
01    image_path = None
02    if image_path is None:
03        print("No image")
04    title = ""
05    if not title:
06        print("Blank title")
07    print(image_path == title)
```

PRO TIP Use is None for the missing-value check.

Run and explain example 4

EXPECTED RESULT

No image
Blank title
False

WHY IT BEHAVES THIS WAY

None and an empty string are different values, although both are falsy.

Exercise 1: Publishing decision

YOUR TASK

Allow publishing only when signed in and a trimmed title has 1 to 160 characters.

CHECK THESE CASES

Signed out fails. Whitespace title fails. " Hello " succeeds when signed in.

HINT AFTER A TRY

Combine the two checks with and.

Exercise 2: Image size decision

YOUR TASK

Classify an upload as missing, acceptable or too large. Limit it to 5 MiB.

CHECK THESE CASES

None is missing. Exactly 5 MiB is acceptable. One byte more is too large.

HINT AFTER A TRY

Check None before comparing numbers.

Checkpoint before moving on

01 Run and explain all four worked examples. Match their actual results to the expected behavior.

02 Attempt both exercises before reading the solutions. Verify every acceptance case.

03 Explain how this chapter supports Postlight: Reject blank titles and allow authors to edit their own posts.

Official reference